

Thermal-mechanical coupling constitutive theory for nonlinear ceramic matrix composite laminates

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Abstract

Temperature-dependent constitutive models under complex stress states are essential for designing and evaluating thermal structures. This study develops an inelastic theoretical framework to characterize the deformation behavior of damageable nonlinear ceramic matrix composites (CMCs) laminates under thermo-mechanical loads, addressing limitations of classical laminate theory (CLT). An anisotropic nonlinear stress-strain model for a single layer is formulated based on post-loading linearization, with damaged compliance and stiffness matrices described via a decoupled damage evolution law. The thermo-mechanical deformation of damaged laminates is expressed using mid-plane strain and curvature, partitioned into elastic and inelastic components. CLT is modified to capture post-damage elastic deformation, leading to a generalized physical equation relating internal forces and moments to mid-plane strain and curvature. For cross-woven CMC laminates, explicit relations are provided. The proposed theory is validated through simulations of on- and off-axis tensile behavior of a 2D-C/SiC composite laminate at elevated temperature.

Keywords

ceramic matrix composites (CMCs), orthotropy, constitutive theory, damage evolution, laminate theory

Introduction

Continuous fiber reinforced composite laminates are widely used in the field of aerospace and marine industries,^{1–5} with designed structures fabricated by stacking two or more single layers through certain techniques. In this material system, multi-directional (MD) laminates composed of multiple single layers with different fiber laying angles are more applicable and attractive.⁶ However, the constitutive behavior of MD laminates can be linear-elastic or nonlinear-inelastic. When characterizing and predicting the linear-elastic constitutive behavior of composite laminates, the classical laminate theory is usually adopted,^{7–9} while for the prediction and characterization of nonlinear stress-strain relationships, damage mechanics theory and progressive damage simulation methods have shown a wider range of application routes and prospects.^{9–13} Without doubt, solutions for the macroscopic nonlinear stress-strain response caused by microscopic damages are quite complex and challenging.^{14–18}

Such problems have been addressed commonly in damageable brittle matrix composites (BMC), including advanced quasi-brittle ceramic matrix composites (CMCs) such as plain-woven C/SiC^{17,19,20} and SiC/SiC^{14,15} laminates, and cross-ply and multidirectional CMC laminates.^{21–23} The nonlinear constitutive behavior of these materials is exactly due to the

microscopic damages, such as matrix cracking, fiber-matrix interface debonding and fiber breakage,^{23–30} caused by applied mechanical and/or thermal loads.^{14–17} Although many studies have been devoted, the damage evolution analysis and the deformation characteristics prediction of laminated CMCs and the bearing performance evaluation of their plate and shell structures still face many difficulties. It is necessary to build an inelastic mechanical-thermal coupling theory framework that correlates macroscopic performance with microscopic damages through advanced cross-scale approaches.³¹

To overcome the linear elasticity and damage independence of the classical constitutive equation and the laminate theory, and to solve the problem of evaluating the

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Data Availability Statement included at the end of the article.

temperature-dependent response performance of vulnerable-nonlinear CMC laminates, this paper will first propose a thermal-mechanical coupling equation based on the damage decoupling conception for anisotropic damage evolution laws, including damage enhancement and deactivation coefficients. Then, for a single layer, an anisotropic nonlinear constitutive relationship equation as function of damage variables and temperature will be suggested. Furthermore, the classical laminate theory will be improved and extended to make it applicable to the characterization and simulation of nonlinear constitutive relationships under thermal-mechanical coupling conditions.

Constitutive model for single layer

Considering plane stress state of an orthotropic layer, denote its in-plane engineering strain components as $\varepsilon_1, \varepsilon_2$ and ε_6 (i.e. γ_{12}) in principal material directions, while the corresponding stress components as σ_1, σ_2 and σ_6 (i.e. τ_{12}). For a CMC layer, it usually presents “post-loading linearization phenomenon”, which means that, after loading-unloading procedure, the reloading stress-strain curve up to the maximum unloading stress is approximately linear. Therefore, the strain-stress relationship in the principal material direction coordinate system can be formulated based on the strain partitioning conception and approach shown in Figure 1, as

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_6 \end{bmatrix} = \begin{bmatrix} \varepsilon_1^e \\ \varepsilon_2^e \\ \varepsilon_6^e \end{bmatrix} + \begin{bmatrix} \varepsilon_1^r \\ \varepsilon_2^r \\ \varepsilon_6^r \end{bmatrix} + \begin{bmatrix} \varepsilon_1^T \\ \varepsilon_2^T \\ \varepsilon_6^T \end{bmatrix} \quad (1)$$

$$= \mathbf{S} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{bmatrix} + \begin{bmatrix} \varepsilon_1^r \\ \varepsilon_2^r \\ \varepsilon_6^r \end{bmatrix} + \begin{bmatrix} \alpha_1(T)T - \alpha_1(T_0)T_0 \\ \alpha_2(T)T - \alpha_2(T_0)T_0 \\ 0 \end{bmatrix}$$

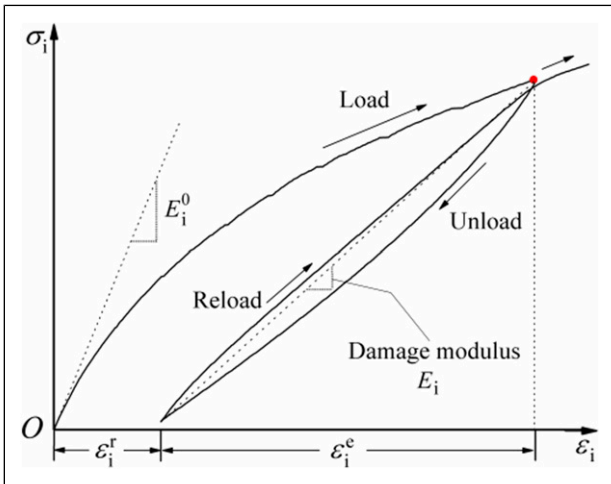


Figure 1. Schematic diagram of strain partitioning based on post-loading linearization phenomenon.

where $\mathbf{S}$ is the compliance matrix of the damaged lamina; α_1 and α_2 are the damage-dependent thermal expansion coefficients in 1-axis and 2-axis, respectively; T is the operating temperature and T_0 is the initial temperature. The superscript “e” represents the elastic part, “r” the inelastic part, and “T” the thermal expansion part, and the same applies below.

There is a simple relationship between the compliance coefficients of a damaged layer and its engineering elastic constants for equation (1). The compliance matrix $\mathbf{S}$ can be expressed as:

$$\mathbf{S} = \begin{bmatrix} \frac{1}{E_1(T)} & -\frac{\nu_{12}(T)}{E_1(T)} & 0 \\ -\frac{\nu_{21}(T)}{E_2(T)} & \frac{1}{E_2(T)} & 0 \\ 0 & 0 & \frac{1}{E_6(T)} \end{bmatrix} \quad (2)$$

In equation (2), E_1, E_2 and E_6 are respectively the damaged moduli in the principal material directions of 1, 2 and 6, and ν_{12} and ν_{21} are respectively the main and secondary damaged Poisson’s ratios. Considering the damage coupling and stress interaction effects, the correlation between the damaged modulus and the effective damage variable is given as follows:

$$E_i(T) = (1 - d_i)E_i^0(T), i = 1, 2, 6 \quad (3)$$

In equation (3), the superscript “0” represents the initial state and E_6 is equivalent to G_{12} . The characterization of the effective damage factor must take into account the coupling effect among anisotropic damages and the damage enhancement/passivation effect caused by the synergistic effect of stress components. The specific expressions for the effective damage variables are^{24,32,33}

$$\begin{bmatrix} d_1 \\ d_2 \\ d_6 \end{bmatrix} = \begin{bmatrix} \omega_{11}(\sigma_1) & \omega_{12}(\sigma_2) & \omega_{16}(\sigma_6) \\ \omega_{21}(\sigma_1) & \omega_{22}(\sigma_2) & \omega_{26}(\sigma_6) \\ \omega_{61}(\sigma_1) & \omega_{62}(\sigma_2) & \omega_{66}(\sigma_6) \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ D_6 \end{bmatrix} \quad (4)$$

$$+ \begin{bmatrix} \lambda_{11}(\sigma_1) & \lambda_{12}(\sigma_2) & \lambda_{16}(\sigma_6) \\ \lambda_{21}(\sigma_1) & \lambda_{22}(\sigma_2) & \lambda_{26}(\sigma_6) \\ \lambda_{61}(\sigma_1) & \lambda_{62}(\sigma_2) & \lambda_{66}(\sigma_6) \end{bmatrix} \begin{bmatrix} \Sigma_1 \\ \Sigma_2 \\ \Sigma_6 \end{bmatrix}$$

where $\omega_{ij}(\sigma_j)$ is the component of damage decoupling coefficients reflecting the mutual influence between the damages in different directions, $\lambda_{ij}(\sigma_j)$ is the component of damage enhancement or deactivation coefficients characterizing mutual promotions in damage between tensile and shear stresses or the inhibitory effect of compressive stress on shear damage; D_i is the principal damage factor under uniaxial mechanical loading at temperature T , and Σ_i is the normalized stress parameter and defined as

$$\Sigma_i = \frac{\sigma_i}{S_i(T)}, (i = 1, 2, 6) \quad (5)$$

where S_i is the temperature-dependent strength component of single layer. Noted that the method for the identification of $\omega_{ij}(\sigma_j)$ and $\lambda_{ij}(\sigma_j)$ at high temperature is the same as at room temperature, i.e., through experimental data fitting, theoretical calculation or numerical simulation.^{32,33}

In addition to the representation of elastic strain, the inelastic strain in equation (1) can be expressed by considering the Poisson's effect, as

$$\begin{bmatrix} \varepsilon_1^r \\ \varepsilon_2^r \\ \varepsilon_6^r \end{bmatrix} = \begin{bmatrix} 1 & -\nu_{21}(T) & 0 \\ -\nu_{12}(T) & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_1^r(d_1) \\ \varepsilon_2^r(d_2) \\ \varepsilon_6^r(d_6) \end{bmatrix} \quad (6)$$

Based on this formula, the residual strain iteration and updating under complex stress can be realized.

In the reference or global coordinate system, the strain-stress relationship of a single layer can be described as

$$\begin{aligned} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} &= \begin{bmatrix} \varepsilon_x^e \\ \varepsilon_y^e \\ \gamma_{xy}^e \end{bmatrix} + \begin{bmatrix} \varepsilon_x^r \\ \varepsilon_y^r \\ \gamma_{xy}^r \end{bmatrix} + \begin{bmatrix} \varepsilon_x^T \\ \varepsilon_y^T \\ \gamma_{xy}^T \end{bmatrix} \\ &= \bar{\mathbf{S}} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} + \begin{bmatrix} \varepsilon_x^r \\ \varepsilon_y^r \\ \gamma_{xy}^r \end{bmatrix} + \begin{bmatrix} \alpha_x(T)T - \alpha_x(T_0)T_0 \\ \alpha_y(T)T - \alpha_y(T_0)T_0 \\ \alpha_{xy}(T)T - \alpha_{xy}(T_0)T_0 \end{bmatrix} \end{aligned} \quad (7)$$

where ε_x , ε_y and γ_{xy} are engineering strain components; σ_x , σ_y and τ_{xy} are stress components; α_x , α_y and α_{xy} are the temperature-dependent thermal expansion coefficients in the off-axis directions. The damage compliance matrix in the reference coordinate system can be expressed as:

$$\bar{\mathbf{S}} = \begin{bmatrix} \bar{S}_{11} & \bar{S}_{12} & \bar{S}_{16} \\ \bar{S}_{12} & \bar{S}_{22} & \bar{S}_{26} \\ \bar{S}_{16} & \bar{S}_{26} & \bar{S}_{66} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_y} & \frac{\eta_{x,xy}}{G_{xy}} \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & \frac{\eta_{y,xy}}{G_{xy}} \\ \frac{\eta_{xy,x}}{E_x} & \frac{\eta_{xy,y}}{E_y} & \frac{1}{G_{xy}} \end{bmatrix} \quad (8)$$

where the temperature-dependent E_x , E_y and G_{xy} are the off-axis stiffness components, ν_{xy} and ν_{yx} are the off-axis Poisson's ratios and $\eta_{x,xy}$, $\eta_{y,xy}$, $\eta_{xy,x}$ and $\eta_{xy,y}$ are the shear-tension and tension-shear coupling coefficients, of the damaged single layer.

Introduce the coordinate transformation matrix $\boldsymbol{\eta}$ for an angle layer, that is

$$\boldsymbol{\eta} = \begin{bmatrix} \cos^2 \theta & \sin^2 \theta & 2 \sin \theta \cos \theta \\ \sin^2 \theta & \cos^2 \theta & -2 \sin \theta \cos \theta \\ -\sin \theta \cos \theta & \sin \theta \cos \theta & \cos^2 \theta - \sin^2 \theta \end{bmatrix} \quad (9)$$

where θ is the off-axis angle. Then the stress and strain components relative to the principal material directions of the single layer can be given by the stress and strain components in the reference system respectively, and the transformation relationship is:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \boldsymbol{\eta} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}, \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix} = (\boldsymbol{\eta}^{-1})^T \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} \quad (10)$$

According to equations (1), and (7) can be transformed to get:

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = \boldsymbol{\eta}^T \mathbf{S} \boldsymbol{\eta} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} + \boldsymbol{\eta}^T \begin{bmatrix} \varepsilon_1^r + \varepsilon_1^T \\ \varepsilon_2^r + \varepsilon_2^T \\ \gamma_{12}^r + \gamma_{12}^T \end{bmatrix} \quad (11)$$

By comparing equation (7) with equation (11), we get the following relationships:

$$\bar{\mathbf{S}} = \boldsymbol{\eta}^T \mathbf{S} \boldsymbol{\eta}, \begin{bmatrix} \varepsilon_x^r \\ \varepsilon_y^r \\ \gamma_{xy}^r \end{bmatrix} = \boldsymbol{\eta}^T \begin{bmatrix} \varepsilon_1^r \\ \varepsilon_2^r \\ \gamma_{12}^r \end{bmatrix}, \begin{bmatrix} \varepsilon_x^T \\ \varepsilon_y^T \\ \gamma_{xy}^T \end{bmatrix} = \boldsymbol{\eta}^T \begin{bmatrix} \varepsilon_1^T \\ \varepsilon_2^T \\ \gamma_{12}^T \end{bmatrix} \quad (12)$$

The conversion relationships shown in equation (12) are simple and effective for characterizing the off-axis stress-strain behavior of a single layer, where the inelastic strain and thermal strain can be obtained through experimental testing or theoretical analysis.

Constitutive theory for CMC laminates

Assume that only intralaminar damages such as matrix cracking, interface debonding and fiber breakage occur in the laminate when subjected to mechanical and thermal loads, meanwhile the layers are firmly bonded to each other and there is no interlaminar damage, i.e., delamination; the total number of layers is n , and the total thickness is h . In the global coordinate system, as shown in Figure 2, the total strain at any point of the nonlinear composite laminate can be expressed as the superposition of elastic strain, inelastic strain and thermal expansion/contraction strain, that is,

$$\begin{aligned} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} &= \begin{bmatrix} \varepsilon_{0x} \\ \varepsilon_{0y} \\ \gamma_{0xy} \end{bmatrix} + z \begin{bmatrix} k_x \\ k_y \\ k_{xy} \end{bmatrix} \\ &= \begin{bmatrix} \varepsilon_{0x}^e + \varepsilon_{0x}^r + \varepsilon_{0x}^T \\ \varepsilon_{0y}^e + \varepsilon_{0y}^r + \varepsilon_{0y}^T \\ \gamma_{0xy}^e + \gamma_{0xy}^r + \gamma_{0xy}^T \end{bmatrix} + z \begin{bmatrix} k_x^e + k_x^r + k_x^T \\ k_y^e + k_y^r + k_y^T \\ k_{xy}^e + k_{xy}^r + k_{xy}^T \end{bmatrix} \end{aligned} \quad (13)$$

where ε_{0x} , ε_{0y} and γ_{0xy} are the mid-plane strain components, and k_x , k_y and k_{xy} are the mid-plane curvature components.

Equation (13) can be abbreviated as:

$$[\epsilon]_{x,y} = [\epsilon_0^e]_{x,y} + [\epsilon_0^r]_{x,y} + [\epsilon_0^T]_{x,y} + z([\kappa^e]_{x,y} + [\kappa^r]_{x,y} + [\kappa^T]_{x,y}) \quad (14)$$

The assumptions of this equation are the same as those in the classical laminate theory, that is, the strain varies linearly along the through-thickness direction of the laminated plate.

According to equation (11), for the k th layer in the laminate, the anisotropic strain-stress relationship is:

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{bmatrix}_k = \begin{bmatrix} \bar{S}_{11} & \bar{S}_{12} & \bar{S}_{16} \\ \bar{S}_{12} & \bar{S}_{22} & \bar{S}_{26} \\ \bar{S}_{16} & \bar{S}_{26} & \bar{S}_{66} \end{bmatrix}_k \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k + \begin{bmatrix} \epsilon_x^r + \epsilon_x^T \\ \epsilon_y^r + \epsilon_y^T \\ \gamma_{xy}^r + \gamma_{xy}^T \end{bmatrix}_k \quad (15)$$

with the off-axis thermal strain components being expressed by

$$\begin{bmatrix} \epsilon_x^T \\ \epsilon_y^T \\ \gamma_{xy}^T \end{bmatrix}_k = \eta_k^T \begin{bmatrix} \epsilon_1^T \\ \epsilon_2^T \\ \gamma_{12}^T \end{bmatrix}_k = \eta_k^T \begin{bmatrix} \alpha_1(T)T - \alpha_1(T_0)T_0 \\ \alpha_2(T)T - \alpha_2(T_0)T_0 \\ 0 \end{bmatrix}_k \quad (16)$$

Assuming that the basic properties of each single layer in the laminate are the same, the damaged compliance matrix of any single layer is:

$$\bar{S}_k = \eta_k^T S_k \eta_k = \eta_k^T \begin{bmatrix} \frac{1}{E_1(T)} & -\frac{\nu_{12}(T)}{E_1(T)} & 0 \\ -\frac{\nu_{21}(T)}{E_2(T)} & \frac{1}{E_2(T)} & 0 \\ 0 & 0 & \frac{1}{E_6(T)} \end{bmatrix}_k \eta_k \quad (17)$$

In equation (17), the coordinate transformation matrix η_k is related to the fiber angle θ of the layer, which is given by equation (9); the damaged compliance matrix S_k is related to the effective damage factor of the layer, as shown in equation (2) and (3).

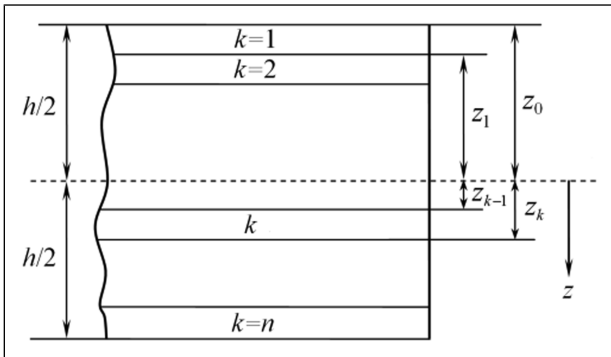


Figure 2. Schematic of geometric laminate model.

Elastic deformation by mechanical load

For the stress of the k th layer of the laminate, by combining equation (12) and (13), we can see that it is related to the damaged stiffness and elastic strain of the single layer, and can be expressed by the mid-surface strain and curvature of the laminate, as

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \left(\begin{bmatrix} \epsilon_{0x}^e \\ \epsilon_{0y}^e \\ \gamma_{0xy}^e \end{bmatrix} + z \begin{bmatrix} \kappa_x^e \\ \kappa_y^e \\ \kappa_{xy}^e \end{bmatrix} \right) \quad (18)$$

Equation (18) can be abbreviated as:

$$[\sigma_{x,y}]_k = \bar{Q}_k [\epsilon_0^e]_{x,y} + z \bar{Q}_k [\kappa^e]_{x,y} \quad (19)$$

where $\bar{Q}_k$ is the inverse matrix of the damaged compliance matrix $\bar{S}_k$, and it is defined as the damaged stiffness matrix of the k th layer, in other words, the reduced stiffness matrix of the damaged single layer.

For the overall laminate, the internal force N and internal moment M can be expressed as the superposition of the internal forces and internal moments of all single layers, that is:

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \sum_{k=1}^n \int_{z_{k-1}}^{z_k} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k dz \quad (20)$$

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \sum_{k=1}^n \int_{z_{k-1}}^{z_k} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k z dz \quad (21)$$

In equations (20) and (21), z_k and z_{k-1} are the coordinate values of the upper and lower surfaces of the k th layer in z -axis, as shown in Figure 2.

By substituting equation (18) into equations (20) and (21), and assuming that the stiffness matrix in each single layer does not change with the coordinate z , and the mid-plane strain and curvature are also independent of the coordinate z , then the relationship between the internal force of the laminate and the mid-plane strain and curvature is obtained as follows:

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \sum_{k=1}^n \left\{ \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \begin{bmatrix} \epsilon_{0x}^e \\ \epsilon_{0y}^e \\ \gamma_{0xy}^e \end{bmatrix} \int_{z_{k-1}}^{z_k} dz \right. \\ \left. + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \begin{bmatrix} \kappa_x^e \\ \kappa_y^e \\ \kappa_{xy}^e \end{bmatrix} \int_{z_{k-1}}^{z_k} z dz \right\} \quad (22)$$

The relationship between the internal moment and the mid-plane strain and curvature is:

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \sum_{k=1}^n \left\{ \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \begin{bmatrix} \epsilon_{0x}^e \\ \epsilon_{0y}^e \\ \gamma_{0xy}^e \end{bmatrix} \int_{z_{k-1}}^{z_k} z dz \right. \\ \left. + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \begin{bmatrix} k_x^e \\ k_y^e \\ k_{xy}^e \end{bmatrix} \int_{z_{k-1}}^{z_k} z^2 dz \right\} \quad (23)$$

Since the mid-plane strain and curvature do not vary with the position of single layer, they can be moved out from the summation. By integrating equations (22) and (23) separately, and noticing the simplified form of equations (14) and (19), the internal force of the nonlinear laminate can be given as:

$$\begin{aligned} N_{x,y} &= \left[\sum_{k=1}^n \bar{Q}_k (z_k - z_{k-1}) \right] [\epsilon_0^e]_{x,y} \\ &+ \left[\frac{1}{2} \sum_{k=1}^n \bar{Q}_k (z_k^2 - z_{k-1}^2) \right] [\kappa^e]_{x,y} \end{aligned} \quad (24)$$

The internal moment is

$$\begin{aligned} M_{x,y} &= \left[\frac{1}{2} \sum_{k=1}^n \bar{Q}^k (z_k^2 - z_{k-1}^2) \right] [\epsilon_0^e]_{x,y} \\ &+ \left[\frac{1}{3} \sum_{k=1}^n \bar{Q}^k (z_k^3 - z_{k-1}^3) \right] [\kappa^e]_{x,y} \end{aligned} \quad (25)$$

The coefficient matrices of elastic strain and curvature in equations (24) and (25) are expressed by $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{D}$ respectively, and are called the damaged in-plane stiffness matrix, damaged coupling stiffness matrix and damaged bending stiffness matrix of the laminate, respectively. These three matrices are symmetric ones, and the components of each stiffness matrix are:

$$\left. \begin{aligned} A_{ij} &= \sum_{k=1}^n \bar{Q}_{ij}^k (z_k - z_{k-1}) \\ B_{ij} &= \frac{1}{2} \sum_{k=1}^n \bar{Q}_{ij}^k (z_k^2 - z_{k-1}^2) \quad (i, j = 1, 2, 6) \\ D_{ij} &= \frac{1}{3} \sum_{k=1}^n \bar{Q}_{ij}^k (z_k^3 - z_{k-1}^3) \end{aligned} \right\} \quad (26)$$

In the formula, the unit of A_{ij} is N/m, the unit of B_{ij} is N, and the unit of D_{ij} is N·m. Equation (24) can be written in matrix form as:

$$N_{x,y} = \mathbf{A} [\epsilon_0^e]_{x,y} + \mathbf{B} [\kappa^e]_{x,y} \quad (27)$$

And equation (25) can be written as:

$$M_{x,y} = \mathbf{B} [\epsilon_0^e]_{x,y} + \mathbf{D} [\kappa^e]_{x,y} \quad (28)$$

The expanded form of equation (27) is

$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{bmatrix} \epsilon_{0x}^e \\ \epsilon_{0y}^e \\ \gamma_{0xy}^e \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} k_x^e \\ k_y^e \\ k_{xy}^e \end{bmatrix} \quad (29)$$

The expanded form of equation (28) can be written as:

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} \epsilon_{0x}^e \\ \epsilon_{0y}^e \\ \gamma_{0xy}^e \end{bmatrix} + \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} k_x^e \\ k_y^e \\ k_{xy}^e \end{bmatrix} \quad (30)$$

By combining equations (29) and (30), the relationship between the generalized force and the generalized mid-surface elastic deformation of the nonlinear laminate is given as follows:

$$\begin{bmatrix} \mathbf{N} \\ \mathbf{M} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \epsilon_0^e \\ \kappa^e \end{bmatrix} \quad (31)$$

From equation (31), it is shown that for nonlinear multi-directional laminates the in-plane force $\mathbf{N}$ is related to both the mid-plane strain and the curvature due to the presence of the coupled stiffness matrix $\mathbf{B}$. Simultaneously, the internal moment $\mathbf{M}$ is not only related to the mid-plane curvature but also affected by the mid-plane strain. This indicates that multi-directional laminates exhibit tension-bending and tension-torsion couplings in addition to tension-shear and bending-torsion couplings. Such coupling effects greatly increase the complexity of the stress-strain response analysis of the laminates.

Moreover, engineering problems often require solving the problems of calculating deformation with known internal forces and internal moments. Therefore, it is necessary to provide the relationship between generalized deformation and internal force of the laminate, that is, the relationship between the mid-plane strain, curvature, internal forces and internal moments. Using the substitution elimination method to equations (27) and (28) to perform matrix operations, we get:

$$\begin{bmatrix} \epsilon_0^e \\ \kappa^e \end{bmatrix} = \begin{bmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{c} & \mathbf{d} \end{bmatrix} \begin{bmatrix} \mathbf{N} \\ \mathbf{M} \end{bmatrix} \quad (32)$$

where matrices $\mathbf{a}$ and $\mathbf{d}$ are the damaged in-plane compliance matrix and the damaged bending compliance

matrix, respectively, and matrices $\mathbf{b}$ and $\mathbf{c}$ are referred to as the damaged coupling compliance matrices of the laminate, whose expressions are:

$$\left. \begin{aligned} \mathbf{a} &= \mathbf{A}^{-1} - \mathbf{B}^* \mathbf{D}^{*-1} \mathbf{C}^* \\ \mathbf{b} &= \mathbf{B}^* \mathbf{D}^{*-1} \\ \mathbf{c} &= -\mathbf{D}^{*-1} \mathbf{C}^* \\ \mathbf{d} &= \mathbf{D}^{*-1} \end{aligned} \right\} \quad (33)$$

The expression of the matrix with superscript “*” in equation (32) are:

$$\left. \begin{aligned} \mathbf{B}^* &= -\mathbf{A}^{-1} \mathbf{B} \\ \mathbf{C}^* &= \mathbf{B} \mathbf{A}^{-1} \\ \mathbf{D}^* &= \mathbf{D} - \mathbf{B} \mathbf{A}^{-1} \mathbf{B} \end{aligned} \right\} \quad (34)$$

From equation (33), it can be deduced that matrices $\mathbf{a}$ and $\mathbf{d}$ are both symmetric ones. Unlike the compliance matrix $\mathbf{S}$ of a damaged single layer, the compliance matrix of a damaged multi-directional laminate does not have an inverse relationship with the corresponding damage stiffness matrix. In addition, it can be shown that the coupled compliance matrices $\mathbf{b}$ and $\mathbf{c}$ have the following relationship:

$$\mathbf{c} = \mathbf{b}^T \quad (35)$$

Combining the above results, equation (32) can be expanded to:

$$\begin{bmatrix} \varepsilon_{0x}^e \\ \varepsilon_{0y}^e \\ \gamma_{0xy}^e \\ k_x^e \\ k_y^e \\ k_{xy}^e \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{16} & b_{11} & b_{12} & b_{16} \\ a_{12} & a_{22} & a_{26} & b_{21} & b_{22} & b_{26} \\ a_{16} & a_{26} & a_{66} & b_{61} & b_{62} & b_{66} \\ b_{11} & b_{21} & b_{61} & d_{11} & d_{12} & d_{16} \\ b_{12} & b_{22} & b_{62} & d_{12} & d_{22} & d_{26} \\ b_{16} & b_{26} & b_{66} & d_{16} & d_{26} & d_{66} \end{bmatrix} \begin{bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{bmatrix} \quad (36)$$

This is the physical relationship between the generalized elastic deformation and the internal force. Equations (32) and (35) can serve as an important supplement and extension to the classical linear-elastic laminate theory.

As is well known, multi-directional laminates, widely used in engineering structures, are typically symmetric. However, the damage states of the symmetric layers in a symmetric laminate are not necessarily identical, so the coupling compliance matrices $\mathbf{b}$ and $\mathbf{c}$ of a damaged geometrically symmetric laminate are not necessarily zero. This means that tension-bending and tension-torsion coupling effects may still exist, complicating the solution.

Inelastic and thermal deformations

Equations (15)–(36) can only characterize the relationship between the elastic deformation and the applied mechanical

load of the multi-directional laminate. As for the mechanical inelastic strain and thermal expansion/contraction strain, they can be derived based on the deformation consistency among each single layer. Firstly, the equivalent internal force corresponding to the inelastic strain and thermal strain can be given as:

$$\begin{bmatrix} \bar{N}_x \\ \bar{N}_y \\ \bar{N}_{xy} \end{bmatrix} = \sum_{k=1}^n \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \left(\begin{bmatrix} \varepsilon_x^r \\ \varepsilon_y^r \\ \gamma_{xy}^r \end{bmatrix} + \begin{bmatrix} \varepsilon_x^T \\ \varepsilon_y^T \\ \gamma_{xy}^T \end{bmatrix} \right) t_k \quad (37)$$

where t_k is the thickness of a single layer. The corresponding equivalent internal moment is:

$$\begin{bmatrix} \bar{M}_x \\ \bar{M}_y \\ \bar{M}_{xy} \end{bmatrix} = \frac{1}{2} \sum_{k=1}^n \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \left(\begin{bmatrix} \varepsilon_x^r \\ \varepsilon_y^r \\ \gamma_{xy}^r \end{bmatrix} + \begin{bmatrix} \varepsilon_x^T \\ \varepsilon_y^T \\ \gamma_{xy}^T \end{bmatrix} \right) t_k h_k \quad (38)$$

where $h_k = z_k + z_{k+1}$. Since the inelastic strain corresponds to the state of complete unloading of the external force, that is, $\mathbf{N} = \mathbf{0}$ and $\mathbf{M} = \mathbf{0}$, the equivalent physical equation of the damaged laminate becomes:

$$\begin{bmatrix} \varepsilon_{0x}^r + \varepsilon_{0x}^T \\ \varepsilon_{0y}^r + \varepsilon_{0y}^T \\ \gamma_{0xy}^r + \gamma_{0xy}^T \\ k_x^r + k_x^T \\ k_y^r + k_y^T \\ k_{xy}^r + k_{xy}^T \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{16} & b_{11} & b_{12} & b_{16} \\ a_{12} & a_{22} & a_{26} & b_{21} & b_{22} & b_{26} \\ a_{16} & a_{26} & a_{66} & b_{61} & b_{62} & b_{66} \\ b_{11} & b_{21} & b_{61} & d_{11} & d_{12} & d_{16} \\ b_{12} & b_{22} & b_{62} & d_{12} & d_{22} & d_{26} \\ b_{16} & b_{26} & b_{66} & d_{16} & d_{26} & d_{66} \end{bmatrix} \begin{bmatrix} \bar{N}_x \\ \bar{N}_y \\ \bar{N}_{xy} \\ \bar{M}_x \\ \bar{M}_y \\ \bar{M}_{xy} \end{bmatrix} \quad (39)$$

According to equation (39), combined with the analysis and prediction of the compliance matrix of the damaged laminate, the mid-plane inelastic strain and curvature components of the laminate can be calculated.

The summary of the above results gives a generalized relationship between the total deformation and the mechanical and equivalent internal forces and moments, which is

$$\begin{bmatrix} \varepsilon_0 \\ \kappa \end{bmatrix} = \begin{bmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{c} & \mathbf{d} \end{bmatrix} \left(\begin{bmatrix} \mathbf{N} \\ \mathbf{M} \end{bmatrix} + \begin{bmatrix} \bar{\mathbf{N}} \\ \bar{\mathbf{M}} \end{bmatrix} \right) \quad (40)$$

Equation (40) provides with a possible route to solve the nonlinear response of CMC laminates subjected thermo-mechanical loads. It should be noted that if no damage occurs upon loading, equation (40) will degenerate into the classical laminate theory (CLT).

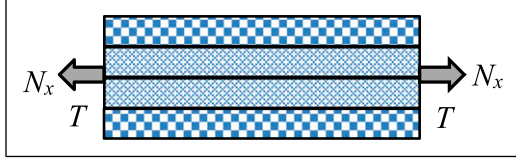


Figure 3. Schematic of cross-woven laminate under thermo-mechanical combined load.

Cross-woven CMC laminate

As an example, a cross-woven CMC laminate is considered here. The cross-woven layer contains only θ and $(\theta-90^\circ)$ fiber yarns, and its mechanical properties are commonly characterized by a $[\theta/\theta-90^\circ]$ laminate model for simplifying analysis while obtaining acceptable results. For a damaged $[\theta/\theta-90^\circ]$ cross-ply laminate, the compliance matrix of single layer relative to the global coordinate system has a simple relationship with its damaged moduli, as shown in equation (8). The damaged compliance matrix $\bar{\mathbf{S}}_\xi$ of the ξ ($\xi = \theta, \theta-90^\circ$) layer is expressed as:

$$\bar{\mathbf{S}}_\xi = \boldsymbol{\eta}_\xi^T \begin{bmatrix} \frac{1}{E_1(T)} & -\frac{\nu_{12}(T)}{E_1(T)} & 0 \\ -\frac{\nu_{21}(T)}{E_2(T)} & \frac{1}{E_2(T)} & 0 \\ 0 & 0 & \frac{1}{E_6(T)} \end{bmatrix} \boldsymbol{\eta}_\xi \quad (41)$$

where $\boldsymbol{\eta}_\xi$ is the coordinate transformation matrix of the ξ layer, its expression is given by equation (9). The damaged stiffness matrix $\bar{\mathbf{Q}}_\xi$ of the ξ layer can be converted from the damaged compliance matrix shown in equation (41) as:

$$\bar{\mathbf{Q}}_\xi = [\bar{\mathbf{S}}_\xi]^{-1} = \begin{bmatrix} \bar{Q}_{11}^\xi & \bar{Q}_{12}^\xi & \bar{Q}_{16}^\xi \\ \bar{Q}_{12}^\xi & \bar{Q}_{22}^\xi & \bar{Q}_{26}^\xi \\ \bar{Q}_{16}^\xi & \bar{Q}_{26}^\xi & \bar{Q}_{66}^\xi \end{bmatrix}, (\xi = \theta, \theta - 90^\circ) \quad (42)$$

It is seen in equation (42) that the tension-shear coupling stiffness coefficients appear, indicating shear deformation will occur under off-axis tension or compression.

According to equation (26), the damaged in-plane stiffness coefficients of the $[\theta/\theta-90^\circ]$ laminate would be:

$$A_{ij} = \sum_\xi t \cdot \bar{Q}_{ij}^\xi = 2t \cdot \bar{Q}_{ij}, (i, j = 1, 2, 6) \quad (43)$$

In equation (43), t means the thickness of single layer, and $\bar{Q}_{ij}$ is referred to as the equivalent stiffness coefficient of the damaged cross-woven layer. Based on equation (43), we have

$$\bar{Q}_{ij} = \frac{1}{2} (\bar{Q}_{ij}^\theta + \bar{Q}_{ij}^{\theta-90^\circ}), (i, j = 1, 2, 6) \quad (44)$$

Equation (44) includes the properties of the cross fiber bundles, and it can be used to characterize the post-damage mechanical behavior of the cross-woven laminates. Such

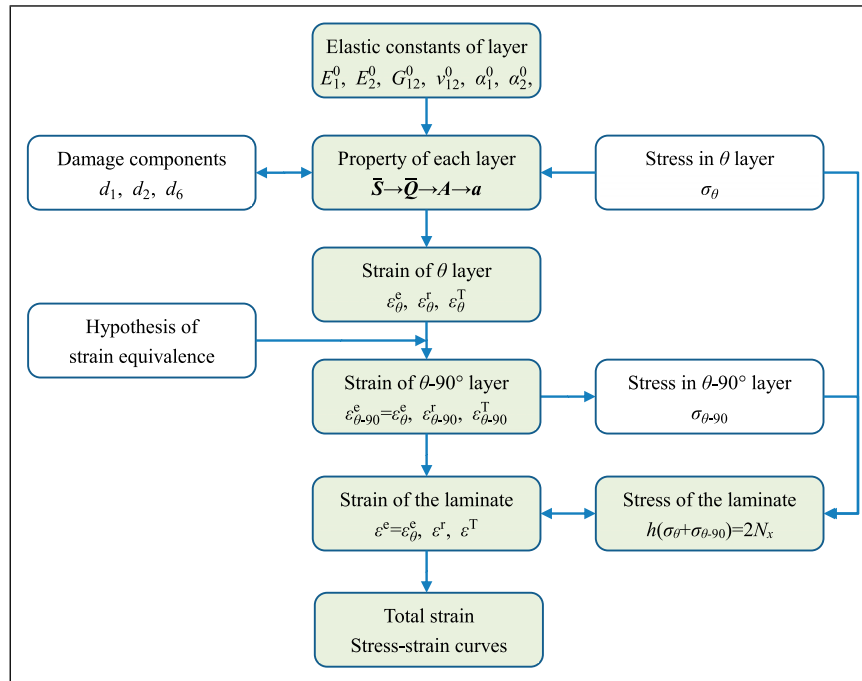


Figure 4. Strain calculation flow chat.

Table 1. Tensile modulus of 2D-C/SiC under different off-radaxis angles and temperatures^{17,36,37}.

Temperature/K	Tensile modulus			
	0°	15°	30°	45°
300	128.7 ± 1.8	78.1 ± 8.3	66.8 ± 11.3	59.6 ± 4.1
973	152.3 ± 4.7	92.6 ± 13.3	84.0 ± 11.5	65.5 ± 14.2
1273	172.7 ± 7.0	122.6 ± 17.2	99.3 ± 15.4	91.1 ± 13.5
1473	169.1 ± 6.2	105.6 ± 4.7	88.5 ± 1.5	79.7 ± 22.7

adaptability is beyond the potential of the classical linear-elastic laminate theory due to occurrence of damage effects.

For the cross-woven laminate $[(\theta/\theta-90^\circ)]_s$ subjected to internal load N_x at temperature T , as shown in Figure 3, the mid-plane strain can be determined by

$$\varepsilon_0 = \mathbf{a} \left(\begin{bmatrix} N_x \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \bar{N}_x \\ \bar{N}_y \\ \bar{N}_{xy} \end{bmatrix} \right) + \mathbf{b} \begin{bmatrix} \bar{M}_x \\ \bar{M}_y \\ \bar{M}_{xy} \end{bmatrix} \quad (45)$$

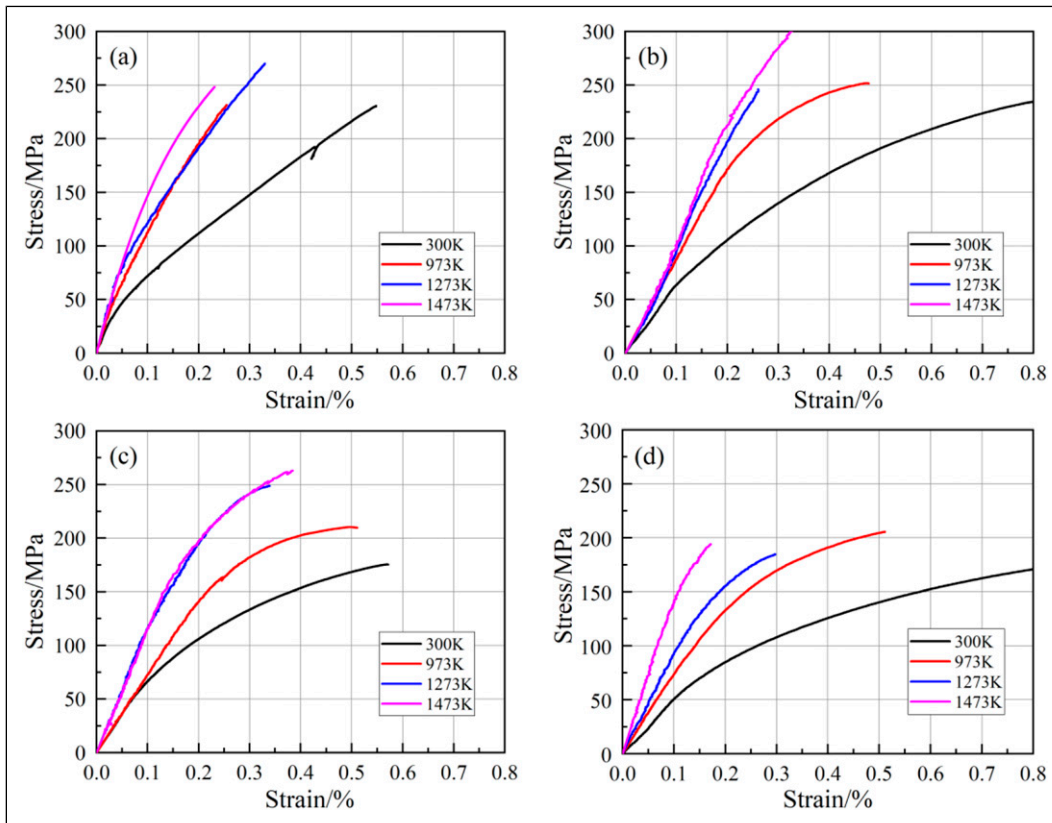
Thermal buckling will occur if there exists thermal gradient. Meanwhile, even though the cross-woven layers own the same composition and structure, their damaged stiffness may be different from each other. Under this case, the stiffness components

A , B and D must be calculated by equation (26), and the compliance matrices $\mathbf{a}$ and $\mathbf{b}$ must be calculated by equation (33) together with equation (34), without concise formulations.

Note here that the procedure how to calculate the strain response of the whole composite laminate is clarified in Figure 4. First of all, it is noticed that the fundamental parameters in equation (41) should be determined. For this purpose, the average stress σ_θ and $\sigma_{\theta-90}$ of the θ and $(\theta-90^\circ)$ layers induced by the internal load N_x is given as

$$\frac{1}{2}(\sigma_\theta + \sigma_{\theta-90}) = \frac{N_x}{h} \quad (46)$$

Consider steady-state thermal environment, then the equivalent internal moment can be neglected for structural

**Figure 5.** Tensile stress-strain curves at elevated temperatures in different loading directions: (a) $\theta = 0^\circ$, (b) $\theta = 15^\circ$, (c) $\theta = 30^\circ$, (d) $\theta = 45^\circ$.

symmetry. The stress components along principal material directions of a single layer are:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{bmatrix}_\xi = \frac{N_x}{h} \begin{bmatrix} \cos^2 \xi \\ \sin^2 \xi \\ -\sin \xi \cos \xi \end{bmatrix} \quad (47)$$

Subjected to the stress σ_θ , the damaged compliance matrix of θ layer can be expressed by using equation (41) or (17) with equation (9), and the damaged stiffness matrix can be obtained by inverting the damaged compliance matrix. Additionally, the elastic strain ε_θ^e can be derived by applying equation (12), while the residual strain ε_θ^r can be given by equation (6). For $(\theta-90^\circ)$ layer, the elastic strain holds $\varepsilon_{\theta-90}^e = \varepsilon_\theta^e$ based on the “deformation coordination hypothesis”, and $\varepsilon_{\theta-90}^e$ can be converted into the strain entity relative to the principal material directions. Subsequently, the corresponding axial damaged modulus of the $(\theta-90^\circ)$ layer can be identified, and its damaged compliance matrix and stiffness matrix can be further determined according to the following basic stress-strain relationship of the layer:

$$\varepsilon_i = \frac{\sigma_i}{E_i} + \varepsilon_i^r + \varepsilon_i^T, i = 1, 2, 6 \quad (48)$$

Based on this calculation, the stress $\sigma_{\theta-90}$ and the residual strain $\varepsilon_{\theta-90}^r$ of the layer can also be obtained.

Since the damaged stiffness matrices $\bar{\mathbf{Q}}_\theta$ and $\bar{\mathbf{Q}}_{\theta-90}$ of the θ and $(\theta-90^\circ)$ layers are obtained, the damaged in-plane

stiffness matrix $\mathbf{A}$ of the laminate can be derived on the basis of equation (26), and the damaged compliance matrix $\mathbf{a}$ can also be get by inverse calculation of $\mathbf{A}$. Then, the mid-surface elastic strain ε_0^e can be calculated according to the physical equation (36). Since the residual strains ε_θ^r and $\varepsilon_{\theta-90}^r$ of the two plies have been obtained already, the overall residual strain ε_0^r of the laminate can be given by equation (39) with the combination of equation (37).

At this point, the overall macroscopic strain ε of the laminate is obtained as the summation of the mid-surface elastic strain ε_0^e and the inelastic strain ε_0^r , and the thermal strain ε_0^T , which can be easily calculated by temperature and damage dependent specific thermal expansion coefficients.^{34,35}

Model application and verification

In practical applications, the developed theory for nonlinear CMC laminates faces two kinds of problems: one is solving deformation with known external thermo-mechanical load, the other is finding external loadings with measured or known deformation. The key task of both problems lies in determination of the damaged compliance and stiffness matrices of single layers and their laminate. The present work focuses on the first problem for model validation, and considers a typical CMC laminate, i.e., two-dimensional cross-woven carbon cloth reinforced silicon carbide composite, namely 2D-C/SiC.

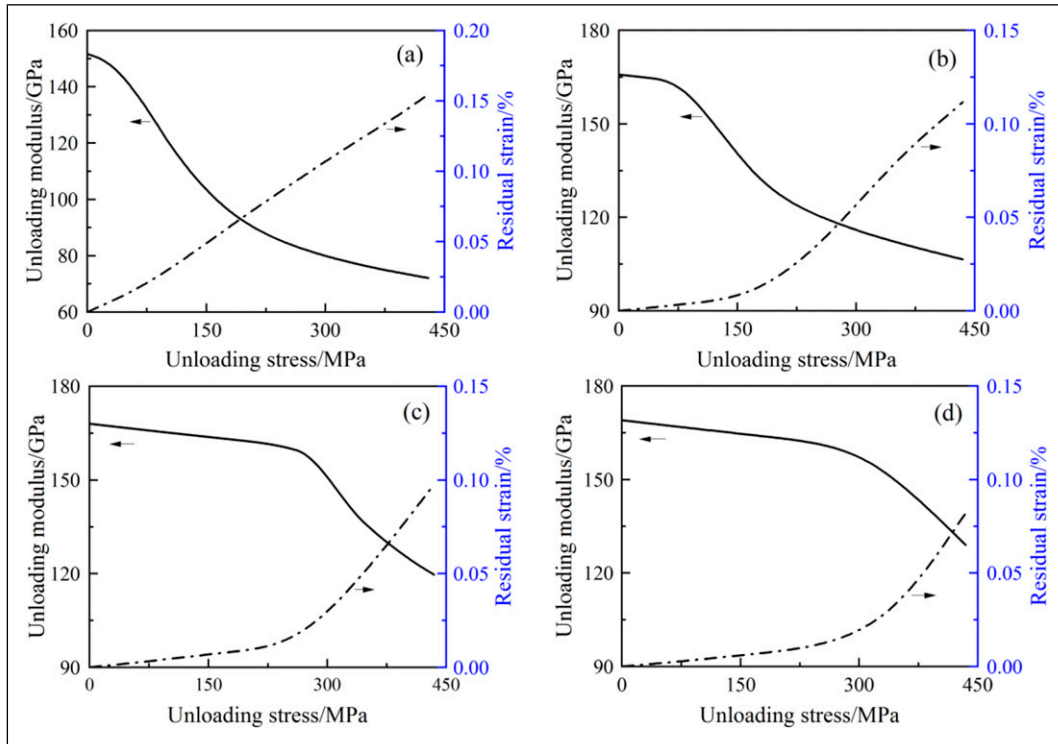


Figure 6. Evolution of unloading modulus and residual strain with unloading stress under on-axis tension at elevated temperature: (a) 300K, (b) 973K, (c) 1273K, (d) 1473K.

The 2D-C/SiC composite material was provided by the National Key Laboratory of Thermostructural Composite Materials at Northwestern Polytechnical University. The material was prepared via chemical vapor infiltration technique with a density of approximately 2.1 g/cm^3 , a fiber volume fraction of about 40%, a theoretical porosity of approximately 16.4%, and a single layer thickness of about 0.2 mm. The configuration of the designed high-temperature tensile specimen is shown elsewhere,^{36,37} with gauge section of 30 mm length, 6 mm width and 3 mm thickness. On- and off-axis tensile tests upon the 2D-C/SiC at elevated temperatures were conducted on a Changchun DNS100 testing machine at a loading speed of 0.3 mm/min, with the longitudinal strain along loading direction measured by an Epsilon 3549 extensometer. The tensile modulus obtained from the nearly linear portion of the tested stress-strain curves at elevated temperatures are listed in Table 1 for a preliminary reference.

Figure 5 shows the typical stress-strain curves of the 2D-C/SiC upon loading in different directions and meanwhile at different temperatures.^{36,37} It can be observed that at the same loading angle and applied stress, strain along the loading direction decreases with increasing temperature, indicating a positive correlation between the apparent stiffness and the ambient temperature. Additionally, the proportional limit,^{38,39} i.e., matrix cracking stress in the

loading direction, tends to increase with rising temperature, while the nonlinearity of the stress-strain curve weakens. It is also noticed that when the off-axis angle increases, the strain values at identical temperature and stress level tend to increase while the off-axis stiffness decreases, concurrently enhancing the nonlinearity of the mechanical behavior. Unfortunately, the thermal expansion deformation was failed to be recorded for the limitation of testing strategy and technology.

The developed constitutive theory is now adapted to simulate the stress-strain behavior of the 2D-C/SiC material under on- and off-axis tension to show its reasonability and applicability, without predicting thermal strains for lack of experimental data. In view of the geometric characteristics of the cross-woven structure of the material, and for the convenience of theoretical modeling, the composite is simplified as $[(\theta/\theta-90^\circ)]_s$ laminate, as shown in Figure 3. It is obvious that the newly developed theory has the potential to calculate the inelastic mechanical response of such symmetrical cross-ply laminate instead of the classical linear-elastic laminate theory, as discussed in the previous section. Being a guideline for model application, one should primarily calculate the compliance S of the 0° layer and then its stiffness Q through inversion of matrix S , and further obtain the compliance and stiffness matrices of -90° , -75° , -60° , -45° , 15° , 30° and 45° layers according to the loading paths of experiment.³⁷

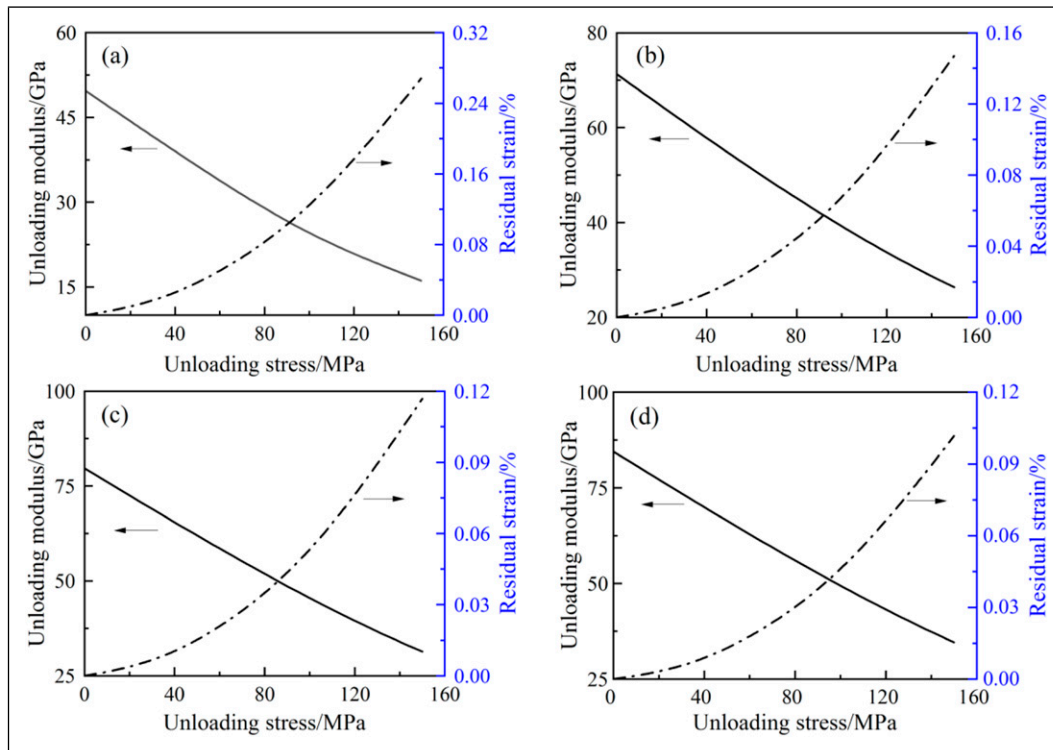


Figure 7. Evolution of unloading modulus and residual strain with unloading stress under in-plane shear at elevated temperature: (a) 300K, (b) 973K, (c) 1273K, (d) 1473K.

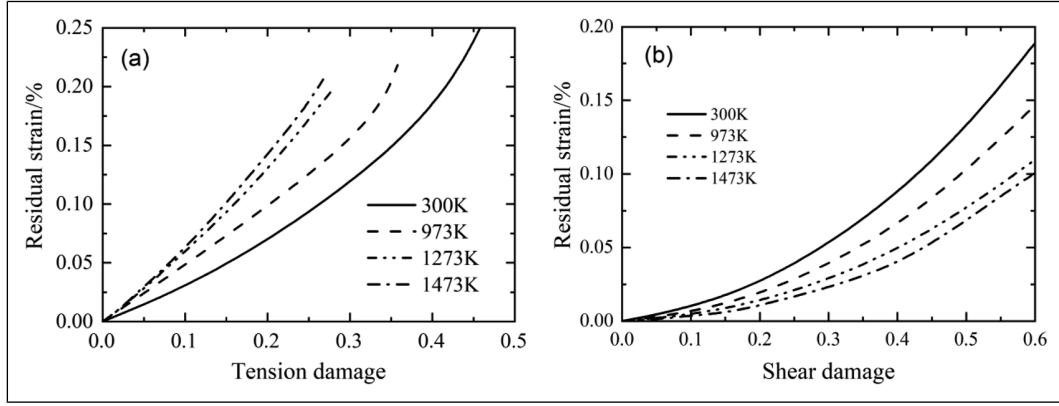


Figure 8. Evolution of residual strain with damage under (a) on-axis tension and (b) in-plane shear at elevated temperatures.

For the 0° layer subjected to on-axis tension or in-plane shear at elevated temperatures, its unloading modulus and residual strain have been studied previously by analytical and theoretical methods^{17,25,36} in accordance with the requirement of equation (2), and the relevant predicted results are provided in Figures 6 and 7 for on-axis tension and in-plane shear scenarios, respectively. It is observed that the unloading modulus continuously declines while

the residual strain grows with increasing stress during the whole loading process under either on-axis tension or in-plane shear. However, such evolution laws are hardly described by concise mathematical expressions. Data can be extracted from Figures 6 and 7 to obtain a basic database for identifying the principal damage components D_i ($i = 1, 2, 6$) and also the residual strain evolution functions, according to:

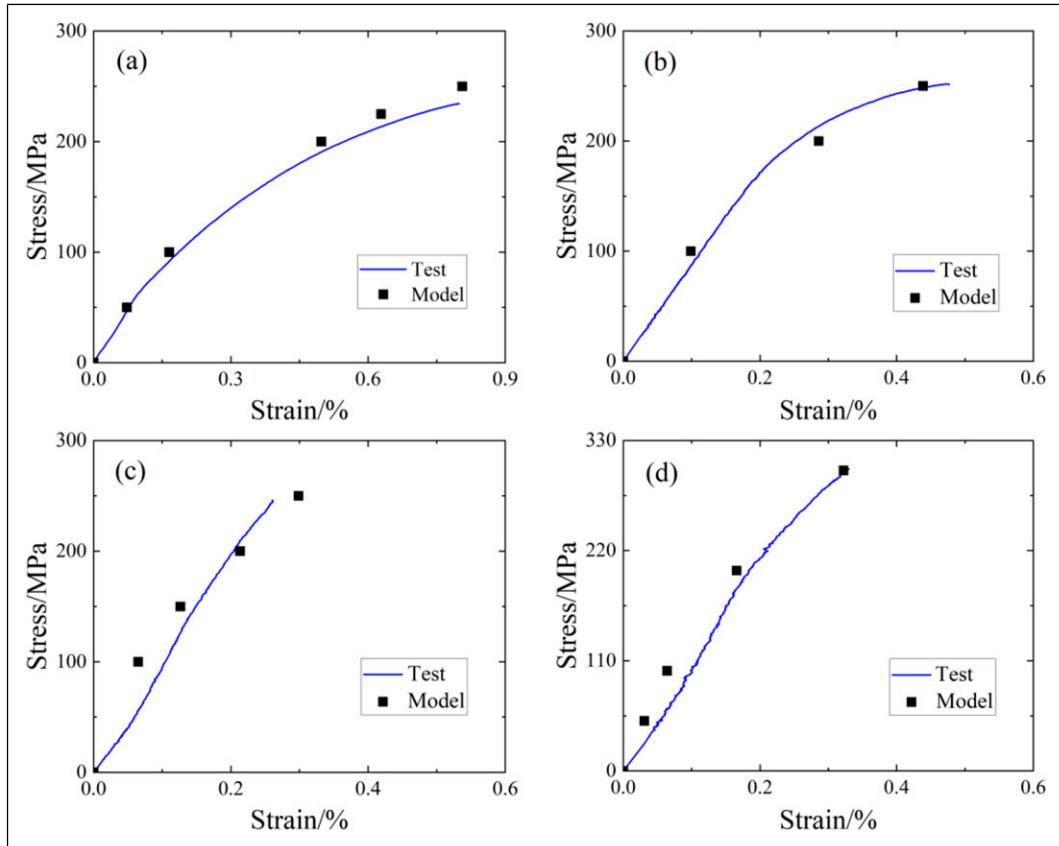


Figure 9. Comparison of experimental and predicted results for 2D-C/SiC composites at $\theta = 15^\circ$: (a) 300K, (b) 973K, (c) 1273K, (d) 1473K.

$$\left. \begin{aligned} D_i &= 1 - E_i/E_i^0 \\ \varepsilon_i^r &= \varepsilon_i^r(\sigma_i) \rightarrow \varepsilon_i^r = \varepsilon_i^r(D_i) \end{aligned} \right\} \quad (49)$$

The damage evolution at the ply-level governs the modulus degradation law and residual strain accumulation of the ply. The ply properties are then transformed into global stiffness and residual strain through laminate-scale expressions. Therefore, both the ply-level damage evolution model and the residual strain model are necessary for the analysis of the overall strain response. The damage evolution model used in this theory can only be formulated based on modulus evolution; damage definitions using other variables would be inapplicable.

Based on the solution of the damage evolution law in terms of unloading modulus, the residual strain component in principal material direction of the 0° layer under either on-axis tension or in-plane shear can be identified as function of damage variable,^{17,25} and the evolution curves are shown in Figure 8. It is observed that the residual strain grows with accumulated damage under either of the two loading cases. Following equation (49), data extracted from Figure 8 can be used for the

characterization of the residual strain evolution laws in terms of damage variable D_i . Such correlation would be very useful if the assumption is reasonable that residual strain depends only on damage in the same direction. Under this hypothesis, the evolution laws can be extended applicable to complex stress state through the following transformation:

$$\varepsilon_i^r = \varepsilon_i^r(D_i) \rightarrow \varepsilon_i^r = \varepsilon_i^r(d_i) \quad (50)$$

Equation (50) is suitable to be used in equation (6), which involves d_i ($i = 1, 2, 6$) accounting for stress interaction effects.

When the laminate subjected to the action of in-plane uniaxial tensile load N_x , each of the bias layers will certainly generate tensile and shear damages, and its damaged moduli in principal material directions can be calculated by equation (3). On this basis, the stiffness A and compliance a can also be obtained following the flow chart shown in Figure 4. During the calculation, the required effective damage factor d_i can be obtained based on the previously identified results from fitting of experimental data, which is formulated as⁴⁰:

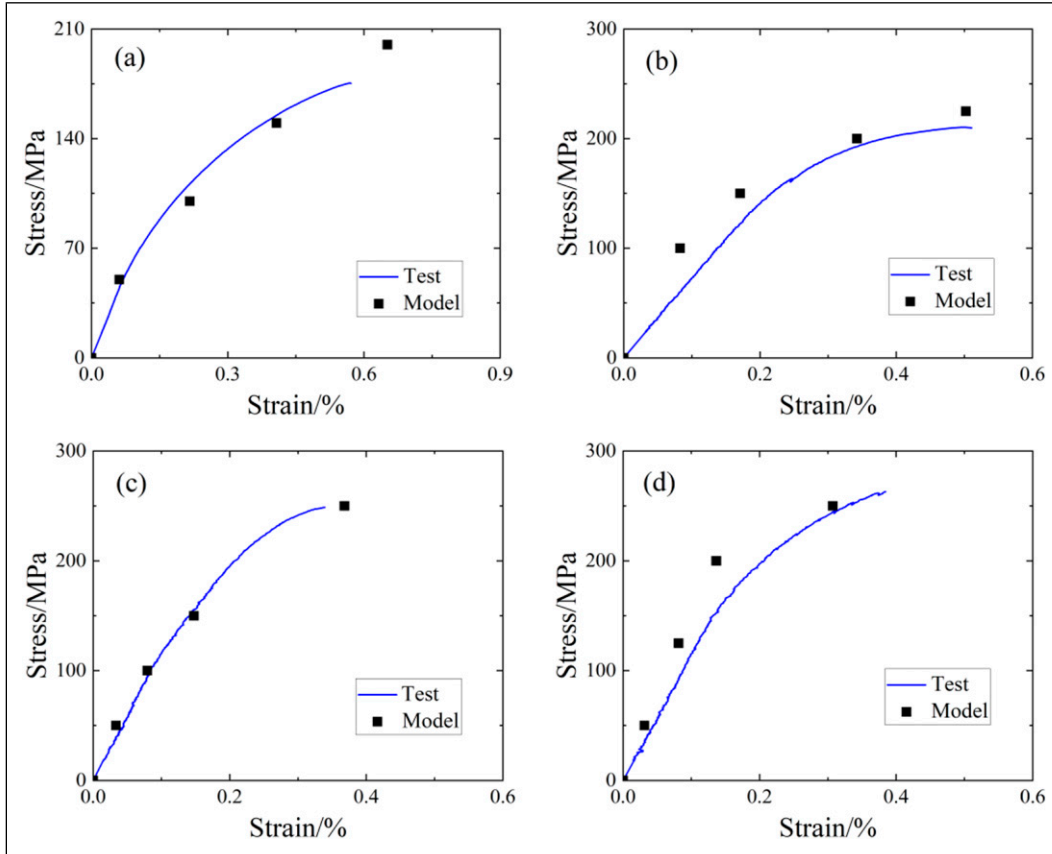


Figure 10. Comparison of experimental and predicted results for 2D-C/SiC composites at $\theta = 30^\circ$: (a) 300K, (b) 973K, (c) 1273K, (d) 1473K.

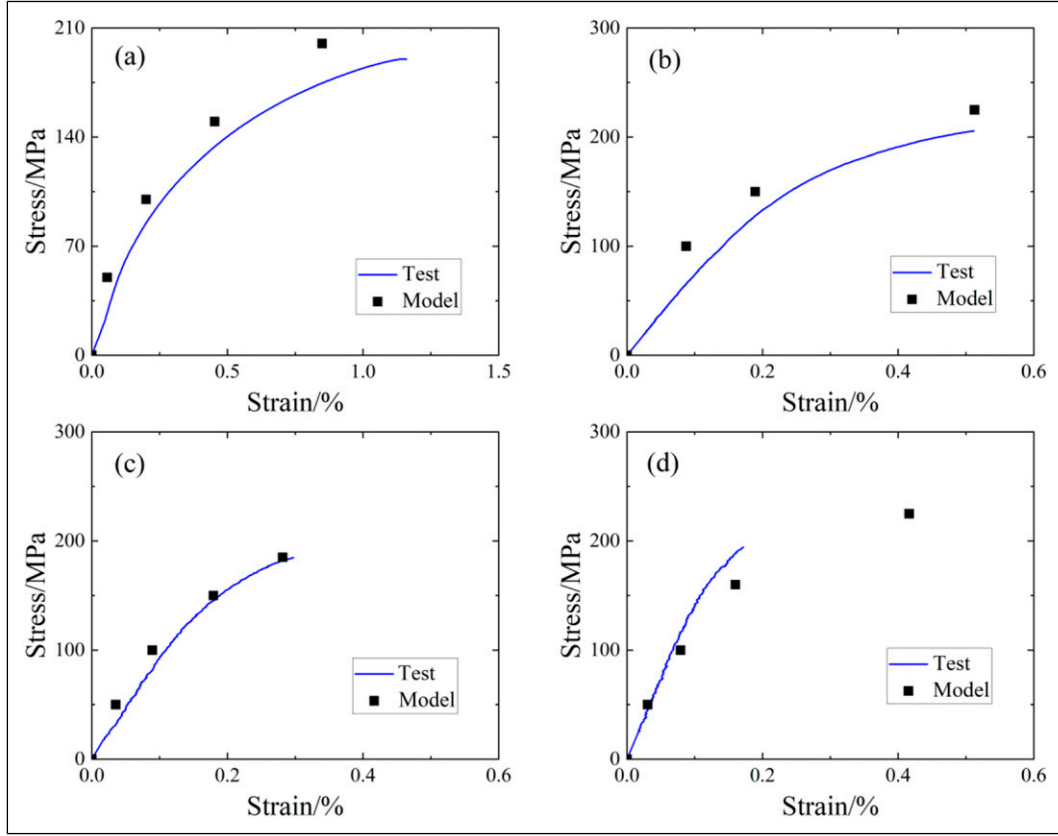


Figure 11. Comparison of experimental and predicted results for 2D-C/SiC composites at $\theta = 45^\circ$: (a) 300K, (b) 973K, (c) 1273K, (d) 1473K.

$$\begin{bmatrix} d_1 \\ d_2 \\ d_6 \end{bmatrix} = \begin{bmatrix} 1 & 1.5 \times 10^{-3} \sigma_2 & 7.8 \times 10^{-5} \sigma_6^2 \\ 1.5 \times 10^{-3} \sigma_1 & 1 & 7.8 \times 10^{-5} \sigma_6 \\ 0.45 & 0.45 & 1 \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ D_6 \end{bmatrix} \quad (51)$$

Equation (51) assumes the damage decoupling coefficients are independent of temperature, because the temperature effect is included in the principal damage components D_i ($i = 1, 2, 6$). Also, it is postulated that the damage coupling modes and mechanisms are the same at elevated temperatures, without oxidation of constituents.

Based on equation (45) and incorporating predicted fundamental property data, the simulated stress-strain curves for 2D-C/SiC laminate under off-axis tension at different temperatures are presented in Figures 9–11. It is observed that the newly proposed model in this paper has strong characterization capabilities for nonlinear behavior. The theoretically predicted curves demonstrate good agreement with experimental results, and the errors can meet the actual engineering requirements, validating the rationality and applicability of the nonlinear laminate theory. The differences between experimental and modeling results may stem from several factors: First, the measured values of high-temperature strain may not be entirely

accurate. Second, the model incorporates numerous influencing factors, leading to significant variability in the predicted values. Third, the multiscale property correlation model itself faces inherent challenges in achieving precise cross-scale property prediction. Even so, the damage evolution laws of the 2D-C/SiC composite material used in this work, i.e., equation (51), still demonstrated its rationality for high-temperature applications.

Conclusion

This paper focuses on mechanical characterization of ceramic matrix composite laminates. Based on the post-loading linearization phenomenon, the classical linear-elastic anisotropic constitutive equation and laminate theory are improved and extended. A damage-based nonlinear constitutive equation and a nonlinear laminate theory are proposed, which incorporate thermal-mechanical coupling effects. A preliminary model verification is carried out by using 2D-C/SiC composite laminates. The main conclusions are drawn as follows.

- (1) The newly proposed nonlinear constitutive relationship model for single layer, which overcomes the drawbacks of classical elastic constitutive

equations, provides a novel route for theoretical modeling of the thermal-mechanical coupling stress-strain behaviors of CMCs.

- (2) The developed nonlinear laminate theory, which is beyond the potential of the classical linear-elastic laminate theory due to consideration of damage effects, provides a simple and effective theoretical approach for analyzing and predicting the constitutive response of nonlinear CMC laminates under complex stress.
- (3) The damage decoupling method and model not only realizes the characterization of damage coupling, enhancement and passivation effects, but also provides a fundamental conception and premise for the accurate description and prediction of damage evolution laws for single layer.
- (4) The nonlinear laminate theory has shown rationality, accuracy and effectiveness in the simulation of stress-strain relationship of plain-woven C/SiC laminated composites, enabling itself having the potential to become an important supplement and development to the theoretical framework of composite laminates.

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Authors contributions

C.-P. Yang: Writing –review & editing, Writing –original draft, Supervision, Resources, Project administration, Funding acquisition. **S.U. Hayat:** Writing –original draft, Writing –review & editing, Investigation, Data curation. **Fei Jia:** Methodology, Conceptualisation, Writing –review & editing. **Weisen Feng:** Writing –review & editing, Validation, Software.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the National Natural Science Foundation of China [grant numbers 12472142, 12072274] and the Fundamental Research Funds for the Central Universities [grant number ZYTS25052].

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

Data will be made available on request.

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